The Prime-Gap Reformulation and the Free Carry

Lean-checked finite algebra for Erdős Problem #251

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Abstract

Let $p_0 = 2, p_1 = 3, \ldots$ enumerate the primes and let $g_n = p_{n+1} - p_n$. We formalise in Lean the exact finite identity

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} \;=\; 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}},$$

with the endpoint term retained, so that it is available before any statement about convergence. For an abstract dyadic tail recurrence $T_{N+1} = 2T_N - g_{N+1}$ with integer digits we prove the block identity $T_{N+h} = 2^h T_N - B_{h,N}$, and hence that the shift $T_{N+h} - T_N$ is integral *if and only if* $(2^h - 1)T_N$ is; that integrality of one shift propagates to every later index; and, by Euler's congruence, that a state with odd reduced denominator d has an integral shift of length $\varphi(d)$.

Problem #251 is open and nothing here decides it. Every statement we prove is finite or algebraic: none of them concerns the infinite series, and connecting the abstract recurrence to the actual prime-gap tail requires a summability argument we do not give. We also record the exact telescoping identity for an unrestricted integer carry, which is the reason a periodic fractional orbit carries no information about periodicity of the gap word; the accompanying construction of prescribed gap patterns is not formalised. The surviving obligation is a growing-block anti-concentration statement for actual consecutive primes.

1 The problem

Let p_n denote the nth prime. Erdős Problem #251 asks whether

$$\Pi = \sum_{n\geq 1} \frac{p_n}{2^n}$$

is irrational [1][2, p. 62][3, p. 103]. Numbering and status follow Bloom's catalogue [4]. The problem is open.

Throughout the formal development the primes are indexed from zero, so that $p_0^{(0)} = 2$, $p_1^{(0)} = 3$, $p_2^{(0)} = 5$, and the gaps are $g_i = p_{i+1}^{(0)} - p_i^{(0)}$. In that indexing

$$\Pi = \sum_{i\geq 0} \frac{p_i^{(0)}}{2^{i+1}},$$

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which is the convention every statement below uses; we drop the superscript from here on. A second normalisation with denominator 2^i also occurs, and at every finite horizon it is exactly twice this one,

$$\sum_{i=0}^{n-1} \frac{p_i}{2^i} = 2\sum_{i=0}^{n-1} \frac{p_i}{2^{i+1}},$$

the factor-of-two normalisation. A factor of two does not change rationality, so no result below depends on the choice; the identity is stated so that the indexing cannot drift silently.

Relation to prior work. Erdős proved that $\sum_n p_n^k/n!$ is irrational for every $k \geq 1$ [1], and conjectured both that $\sum_n p_n^k/2^n$ is irrational for every k and that $\sum_{n\geq 1} p_n/(g_1 \cdots g_n)$ is irrational whenever $g_n \geq 2$ and $g_n = o(p_n)$ [3, p. 103]; the choice $g_n = p_n + 1$ shows that some growth condition is needed there. The statement of Problem #251 has been formalised before, as a conjecture with an unfilled proof, in the *formal-conjectures* collection. No claim of priority is made for anything below.

Two routes and where they stop. The digits p_n grow, so the series is not a digit expansion in any bounded alphabet, and the standard rationality criteria for such expansions do not apply directly. Two rewritings suggest themselves. The first replaces the primes by their gaps, which are bounded on average and are the natural object of prime-distribution theory; Section 2 carries this out exactly, and identifies the endpoint term that a passage to the limit must control. The second reads rationality as a constraint on a tail orbit; Section 3 develops that constraint, and Section 4 gives the exact reason it does not constrain the gaps themselves.

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Reading map. Section 2 is the reformulation, Section 3 the tail algebra, Section 4 the no-go, and Section 5 the remaining obligation. Linked phrases open the corresponding Lean declaration at the pinned source revision 4eb4c8ecfa0a.

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MSC 2020. 11J72 (primary); 11N05, 68V20 (secondary).

2 Summation by parts, with the endpoint retained

Write $\mathbb{N}_{>0}$-indexed sums with the convention

$$D(P,n) = \sum_{i=0}^{n-1} \frac{P(i)}{2^{i+1}}, \qquad \Delta(P,n) = \sum_{i=0}^{n-1} \frac{P(i+1)-P(i)}{2^{i+1}},$$

Statement	Status	Treatment here
Irrationality of Π	Open	Not proved.
Finite prime-gap identity	Proved here	Theorem 2.2, with the endpoint retained.
Infinite prime-gap identity	Not proved	Requires $p_n 2^{-n} \to 0$, which we do not establish.
Block identity for the tail recurrence	Proved here	Theorem 3.2.
Integral shift criterion	Proved here; an equivalence	Theorem 3.3.
Totient shift from an odd denominator	Proved here	Theorem 3.4.
Propagation of an integral shift	Proved here	Theorem 3.5.
Rationality forces periodic gaps	False	Section 4; the carry is free.
Anti-concentration for actual gaps	Open	Section 5; the surviving obligation.

for a rational sequence P. These are the dyadic partial sum and the dyadic difference sum.

Proposition 2.1 (finite summation by parts)**.** *For every rational sequence P and every $n \geq 0$,*

$$D(P, n+1) = P(0) + \Delta(P, n) - \frac{P(n)}{2^{n+1}}.$$

Proof. Induction on n. At $n = 0$ both sides equal $P(0)/2$. For the step, adding $P(n+1)/2^{n+2}$ to the left and $(P(n+1) - P(n))/2^{n+1}$ to the difference sum changes the endpoint term from $P(n)/2^{n+1}$ to $P(n+1)/2^{n+2}$, and the two adjustments agree. □

Formalised as the summation-by-parts identity. Nothing is assumed about P: no positivity, no monotonicity, and no convergence.

Specialising to $P(i) = p_i$, whose first value is $p_0 = 2$, and writing $g_i = p_{i+1} - p_i$ for the zero-based gaps, gives the reformulation.

Theorem 2.2 (prime-gap reformulation)**.** *For every $n \geq 0$,*

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Formalised as the prime-gap summation by parts, using the gap partial sum and the gap cast identity; the latter records that the natural-number difference $p_{n+1} - p_n$ agrees with the difference taken in $\mathbb{Q}$, which needs $p_n \leq p_{n+1}$, the monotonicity step. The leading 2 is the first prime, not a normalising constant.

What the identity does and does not give. The identity is exact at every finite horizon and free of hypotheses. It is not the corresponding statement about infinite series. Passing to the limit requires $p_n 2^{-n} \to 0$, and while that is elementary, it is not proved in the formal source and is therefore not available to any statement here. We record the finite identity in this form deliberately: the endpoint term is the whole content of the passage to the limit, and suppressing it is exactly the step that would make a finite identity look like an infinite one.

3 The tail recurrence and integral shifts

Rationality of a dyadic series constrains its tail orbit. This section proves what that constraint is, for an abstract recurrence.

Definition 3.1. Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{Q}$. Say T satisfies the *dyadic tail recurrence* with digits g when

$$T_{N+1} = 2T_N - g_{N+1} \qquad \text{for every } N.$$

Write $\sigma_h(N) = T_{N+h} - T_N$ for the *shift* of length h at N, and call a rational number *integral* when it is the image of an integer.

These are the tail recurrence, the shift, and integrality. The digits are arbitrary integers. In the intended instance they are the prime gaps and T_N is the scaled tail of Π after level N, but that instantiation needs a summability argument and is not made here; every statement below is a theorem about Definition 3.1.

Iterating the recurrence accumulates an explicit integer. Define $B_{0,N} = 0$ and $B_{h+1,N} = 2B_{h,N} + g_{N+h+1}$, so that $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$: the tail block.

Theorem 3.2 (block identity)**.** *For every N and h,*

$$T_{N+h} = 2^h T_N - B_{h,N}, \qquad \textit{hence} \qquad \sigma_h(N) = (2^h - 1)\, T_N - B_{h,N}.$$

Proof. Induction on h; the step is one application of the recurrence together with $2 \cdot 2^h = 2^{h+1}$ and the recursion defining B. The second identity is the first minus T_N. □

Formalised as the iterated block identity and the scaled shift identity. The shift also obeys the recurrence in its own right, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$, the shift step identity.

Since $B_{h,N}$ is an integer, Theorem 3.2 converts a question about the shift into a question about a single scaled state.

Theorem 3.3 (integral-shift criterion)**.** *For every N and h, the shift $\sigma_h(N)$ is integral if and only if $(2^h - 1)T_N$ is integral.*

Proof. By Theorem 3.2 the two differ by the integer $B_{h,N}$, and subtracting an integer does not change integrality. □

Formalised as the integral-shift criterion, on the integer-shift invariance. This is an equivalence, not a one-way reduction: the block carries no information about integrality, so the two conditions are the same condition.

Theorem 3.4 (a shift of totient length)**.** *If the reduced denominator d of T_N is odd, then $\sigma_{\varphi(d)}(N)$ is integral.*

Proof. Since d is odd, 2 and d are coprime, so Euler's congruence gives $2^{\varphi(d)} \equiv 1 \pmod{d}$, that is $d \mid 2^{\varphi(d)} - 1$. Writing $2^{\varphi(d)} - 1 = dk$ and $T_N = u/d$ in lowest terms, $(2^{\varphi(d)} - 1)T_N = ku$ is an integer, and Theorem 3.3 transfers this to the shift. □

Formalised as the totient shift and the Euler multiplier. The hypothesis is a genuine restriction: the argument uses coprimality of 2 with the denominator, and the even part of a denominator is exactly what the doubling in the recurrence acts on.

Theorem 3.5 (propagation)**.** *If $\sigma_h(N)$ is integral, then $\sigma_h(N+k)$ is integral for every $k \geq 0$.*

Proof. By the shift step identity, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$ is an integer combination of an integer and two digits. Induct on k. □

Formalised as the one-step propagation and the propagation to every later index.

What this section proves, and what it does not. Theorems 3.2–3.5 are statements about Definition 3.1 and hold for arbitrary integer digits. None of them mentions primes, and none of them uses convergence. Instantiating them at the prime-gap tail is a separate step: one must first prove that the scaled tail of Π exists and satisfies the recurrence, and we do not. In particular nothing here shows that Π has, or does not have, an integral shift.

4 The carry is free

A natural hope is that rationality of a dyadic series forces its digit stream to be eventually periodic, or at least automatic, and that this is impossible for prime gaps. The hope fails at the first step, and the reason is a one-line identity.

Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary and put $\kappa_n = 2K_n - K_{n+1}$: the carry coefficient.

Proposition 4.1 (exact telescoping)**.** *For every $n \geq 0$,*

$$\sum_{i=0}^{n-1} \frac{\kappa_i}{2^{i+1}} = K_0 - \frac{K_n}{2^n}.$$

Proof. Induction on n; the added term is $(2K_n - K_{n+1})/2^{n+1} = K_n/2^n - K_{n+1}/2^{n+1}$. □

Formalised as the carry telescoping identity, on the carry partial sum. When the carries are natural numbers and $K_{n+1} \leq 2K_n$, the coefficients are natural numbers and the two readings agree, the natural-carry cast.

The reading, and its exact status. The identity says that the carry sequence K is unconstrained: choose K freely, and the emitted coefficients κ_n have dyadic partial sums telescoping to K_0 up to $K_n 2^{-n}$. A rational limit therefore places no periodicity or automaticity requirement on the coefficient stream, because the same rational value arises from carry sequences that are not eventually periodic. This is why the tail constraint of Section 3 does not descend to the gap word: rationality controls a fractional orbit, and the integer part absorbs everything else.

Proposition 4.1 is proved; the accompanying claim that prescribed finite gap patterns can be realised by such a carry is a construction we have not formalised, and it is not used anywhere above. What is established is the identity and the consequence that the periodicity route is closed, not a classification of which coefficient streams occur.

5 Complements and further questions

Problem #251 is open. The results above fix the shape of what is missing.

Problem 5.1 (growing-block anti-concentration)**.** Show that for the actual sequence of consecutive prime gaps, every sufficiently long block stays a definite dyadic distance from every carry-compatible rational target, with the required length growing with the starting index.

Problem 5.2 (the input from prime distribution)**.** Identify a theorem about consecutive primes strong enough to supply that block obstruction. The occurrence of individual gap values, however often, is not enough: by Section 4 a finite insertion of any fixed even gap word is compatible with a rational value, so the obstruction must be a statement about long blocks rather than about single gaps.

Two smaller obligations remain on the formal side. The infinite form of Theorem 2.2 needs $p_n 2^{-n} \to 0$; and instantiating Definition 3.1 at the prime-gap tail needs summability. Neither is difficult, and neither is done, so no statement above may be read as concerning the infinite series. What the formal content changes is the location of the difficulty: it is not in the reformulation, and it is not in the tail algebra.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 4eb4c8ecfa0a. All declarations of this note live in one module. The prime-specific material is confined to Section 2; everything in Sections 3 and 4 is stated for arbitrary integer digits and arbitrary rational carries, and the reader should not transfer those statements to the primes without the summability step named in Section 5.

References

[1] P. Erdős, *Sur certaines séries à valeur irrationnelle*, Enseign. Math. (2) **4** (1958), 93–100. MR 98732.

[2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 62.

[3] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[4] T. F. Bloom, *Erdős Problem #251*, `erdosproblems.com/251`, accessed 22 July 2026.